

INTRODUCTION TO EMBEDDED SYSTEM

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DEFINITION

- An embedded system is a computer hardware with embedded software as one of its most important components.
- It is a dedicated computer-based system for an application or product.
- It may be either an independent system or a part of a larger system.

EXAMPLE



Router



Smartwatch



Microwave Oven

KEY FEATURES OF EMBEDDED SYSTEMS

- Dedicated Functionality
  - Unlike PC's – NOT General Purpose

- Real Time Operations
  - Respond to changes in input/environment within strict time constraints.

COMPUTER AND EMBEDDED SYSTEM(1)

A computer is a system that has the following or more components.

1. A microprocessor
2. A large memory comprising the following two kinds:
    (a) Primary memory (RAM, ROM and fast accessible caches)
    (b) Secondary memory (HDD, SSD, USB)
3. Input units like keyboard, mouse, digitizer, scanner, etc.
4. Output units like monitor, printer, etc.
5. Networking units like Ethernet card, front-end processor-based drivers, etc.
6. I/O units like a modem, fax, etc.

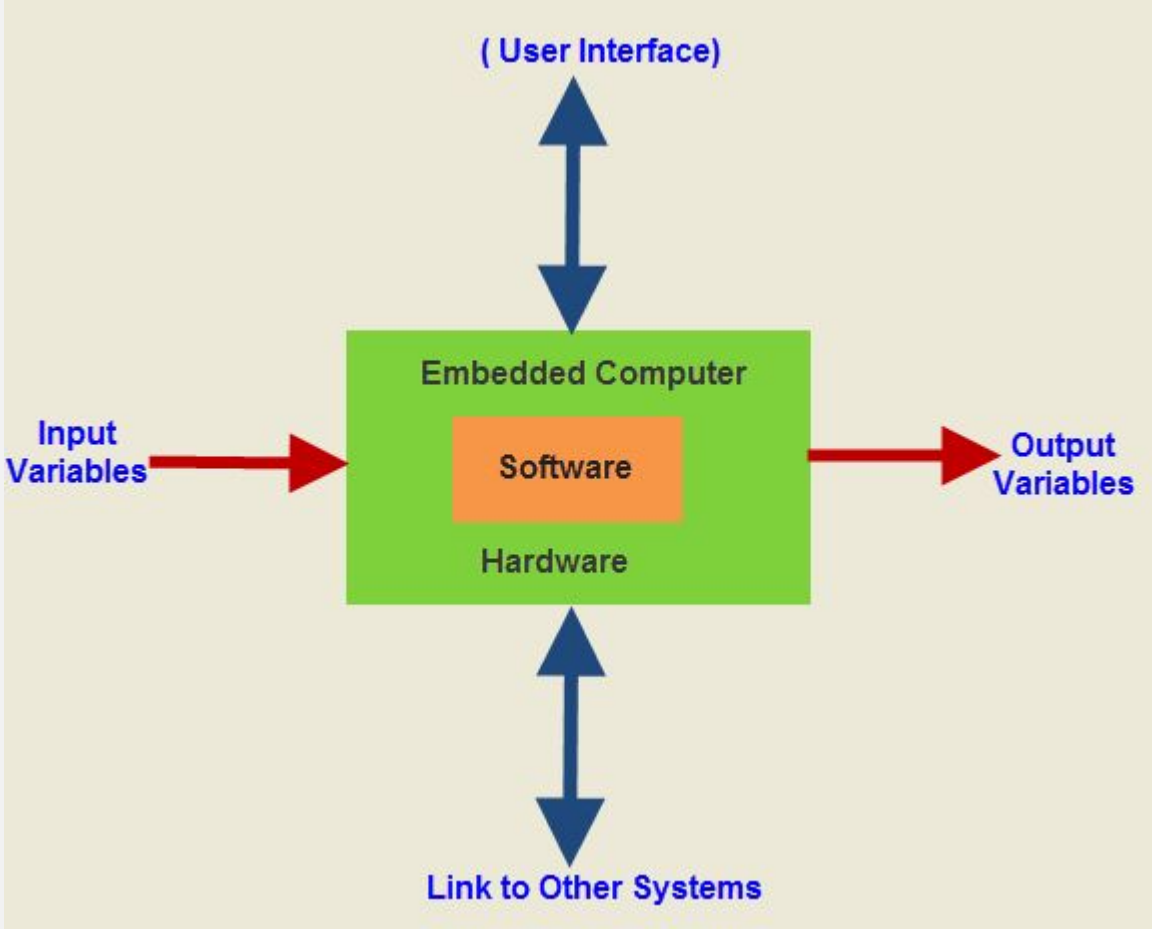
COMPUTER AND EMBEDDED SYSTEM(2)

Incase of embedded system, its software usually embeds in ROM (Read Only Memory). It does not need secondary memories as in a computer.

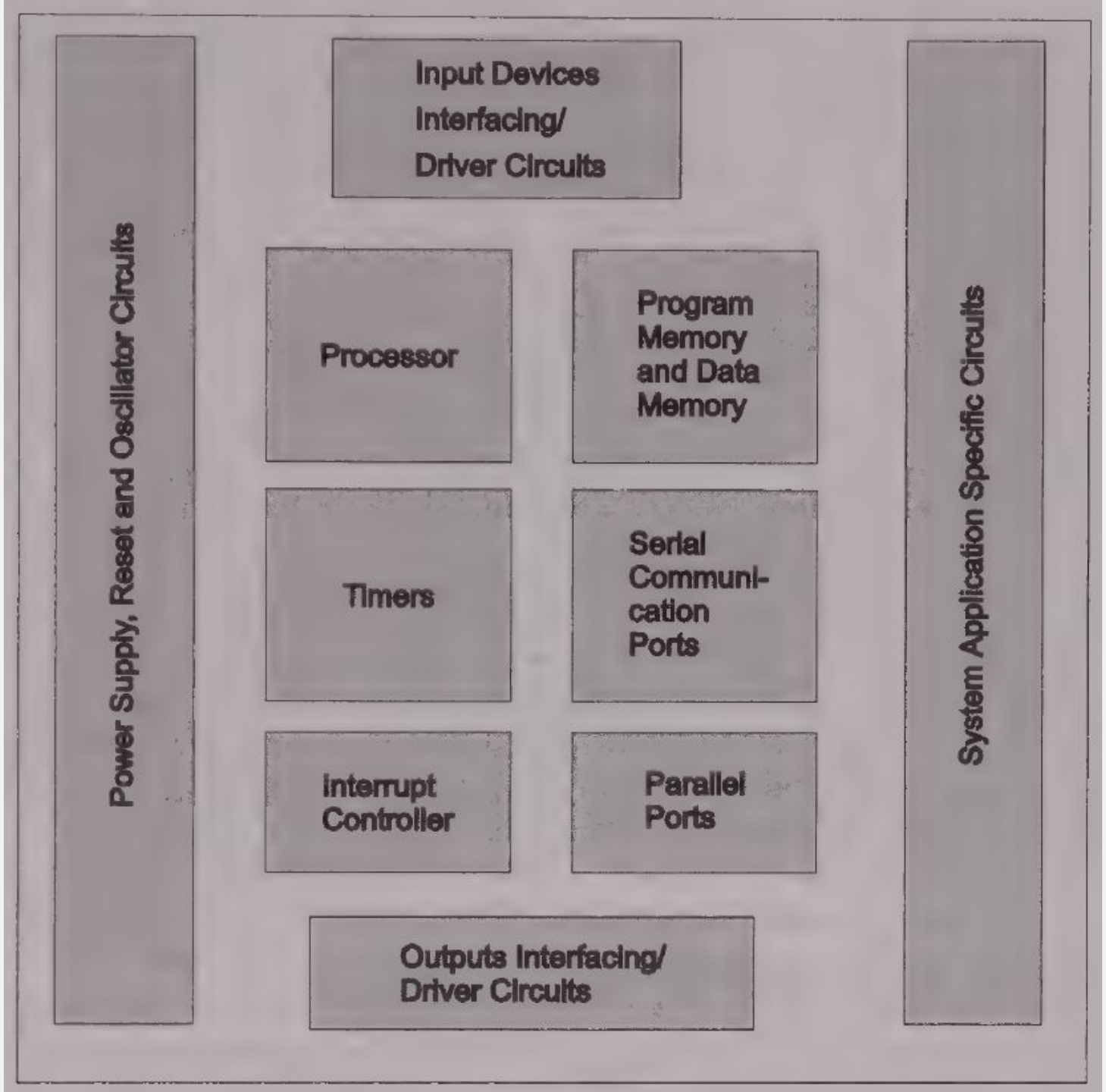
An embedded system has three main components:

1. Hardware
2. Software
3. Firmware (RTOS – Real Time Operating System

BASIC BLOCK DIAGRAM OF EMBEDDED SYSTEM



The components of an embedded system hardware



DIFFERENCE BETWEEN GENERAL PURPOSE COMPUTING SYSTEM & AN EMBEDDED SYSTEM

General Purpose Computing System	Embedded System
A system which is a combination of generic hardware and a general-purpose operating system for executing a variety of applications.	A system which is a combination of special-purpose hardware and an embedded OS for executing a specific set of applications.
Contains a general-purpose operating system.	May or may not contain an OS for functioning.
Applications are alterable by the end user.	The firmware of an embedded system is pre-programmed and is non-alterable by the end user.
Performance is the key deciding factor in the selection of the system.	Application-specific requirements (like power requirements, performance, memory usage, etc.) are key deciding factors.
Need not be deterministic in execution behavior.	Execution behavior is deterministic for certain types of embedded systems, like hard real-time systems.
Examples: Computer, laptop	Examples: DVD player, mobile phones, printers

EXAMPLES

- Digital watches
- Washing Machine
- Toys
- Televisions
- Feature phones
- Laser Printer
- Cameras
- Industrial machines
- Electronic Calculators
- Automobiles
- Medical Equipment

EMBEDDED SYSTEM APPLICATIONS

Application of Embedded System in Day to Day Life Devices:

1. **Washing Machines:** Automatically control washing cycles, water levels, and spin speed.
2. **Microwave Ovens:** Control cooking time, power levels, and heating operations.
3. **Network Routers:** Manage data traffic between connected devices and the internet.
4. **Air Conditioners:** Automatically regulate room temperature based on sensor readings.
5. **Smart TVs:** Control display functions, process user inputs, and run applications.
6. **Automobiles:** Use Electronic Control Units (ECUs) to control engine operation, braking systems, and other vehicle functions.
7. **Digital Cameras:** Control image capture, autofocus, and image processing.
8. **Medical Wearables:** Monitor heart rate, physical activity, and other health measurements.
9. **Smart Door Locks:** Control door locking and unlocking using electronic keys, passwords, or fingerprints.

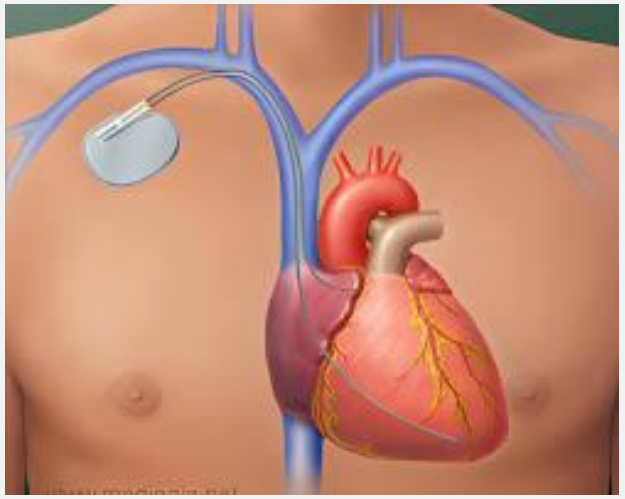
EMBEDDED SYSTEM APPLICATIONS

Application of Embedded Systen in Industrial Control Devices:

- Manage and monitor industrial processes to ensure efficiency and safety.
- Manage energy distribution and optimize grid performance.
- Monitor and report energy usage.

Application of Embedded System in Medical Devices:

- Pacemakers
- Insulin pumps
- Diagnostic equipment
- Imaging machines



Pacemaker

CHARACTERISTICS OF EMBEDDED SYSTEM

- **Dedicated purpose**: Designed to perform specific tasks or functions.
- **Real-time operation**: Often required to respond to inputs within strict time constraints.
- **Resource constraints**: Limited processing power, memory, and energy consumption.
- **Reliability**: Must operate consistently and dependably, often in critical applications.
- **Long-term operation**: Expected to run for extended periods without interruption or maintenance.
- **Integration**: Tightly integrated with physical hardware and often interacts directly with the environment.
- **Customized hardware and software**: Tailored to the specific application requirements.
- **Limited user interface**: Often minimal or no direct user interaction.
- **Reactive and adaptive**: Responds to changes in the environment or system state.
- **Cost-effectiveness**: Designed to be economical for mass production in many cases.

THANK YOU